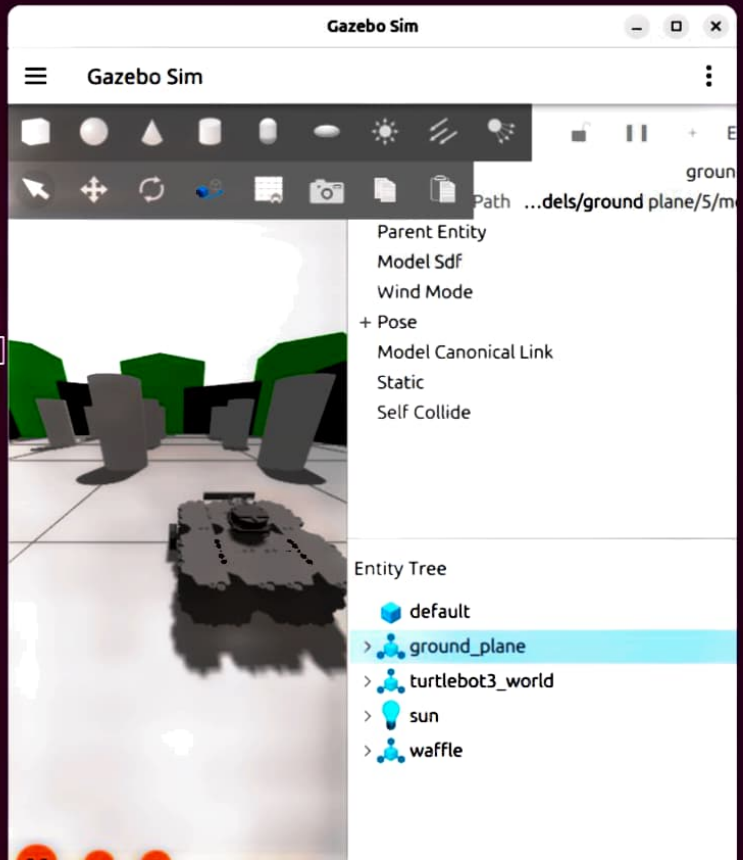
youssefmohamed@youssefmohamed-MacBookPro12-1: ~/autonomous_ws

youssefmohamed@youssefmohamed-MacBookPro12-1: ~/autonomous_ws

```
youssefmohamed@youssefmohamed-MacBookPro12-1:~/autonomous_ws$ ros2 topic list
/camera/camera_info
/camera/image_raw
/camera/image_raw/compressed
/clock
/cmd_vel
/imu
/joint_states
/odom
/parameter_events
/robot_description
/rosout
/scan
/tf
/tf_static
youssefmohamed@youssefmohamed-MacBookPro12-1:~/autonomous_ws$
```



youssefmohamed@youssefmohamed-MacBookPro12-1: ~/autonomous_ws

```
L Button: Binding loop det
[gazebo-2] [GUI] [Wrn] [Ap
L ToolButton: Binding loop
[gazebo-2] [GUI] [Wrn] [Ap
L Button: Binding loop det
[gazebo-2] [GUI] [Wrn] [Ap
L Button: Binding loop det
[gazebo-2] [GUI] [Wrn] [Ap
L ToolButton: Binding loop
[gazebo-2] [GUI] [Wrn] [Ap
L Button: Binding loop det
[gazebo-2] [GUI] [Wrn] [Ap
L Button: Binding loop det
[gazebo-1] [Wrn] [Ogre2Cam
 the reserved bits used in
[gazebo-2] [GUI] [Wrn] [Ap
L ToolButton: Binding loop
[gazebo-2] [GUI] [Wrn] [Ap
L Button: Binding loop det
[gazebo-2] [GUI] [Wrn] [Ap
L Button: Binding loop det
[gazebo-2] [GUI] [Wrn] [Ap
 for property "implicitHei
[gazebo-2] [GUI] [Wrn] [Ap
L ToolButton: Binding loop
[gazebo-2] [GUI] [Wrn] [Ap
L Button: Binding loop det
[gazebo-2] [GUI] [Wrn] [Ap
L Button: Binding loop det
[gazebo-2] Warning [Utils.
<data-string>:L163]: XML E
[gazebo-2] Warning [Utils.
:<data-string>:L394]: XML
.
[gazebo-2] [GUI] [Wrn] [Ap
"x"
```

Gazebo Sim

Gazebo Sim

Model — Entity 4

Name	ground_plane
Source File Path	...dels/ground plane/5/model.sdf
Parent Entity	1
Model Sdf	
Wind Mode	
+ Pose	
Model Canonical Link	
Static	
Self Collide	

Entity Tree

- default
- ground_plane
- turtlebot3_world
- sun
- waffle

```
l:413:17: QM
l:309:21: QM
l:309:21: QM
l:413:17: QM
l:309:21: QM
l:309:21: QM
onflict with
l:413:17: QM
l:309:21: QM
l:309:21: QM
oop detected
l:413:17: QM
l:309:21: QM
l:309:21: QM
gz_frame_id:
of [sensor].
/gz_frame_id
 of [sensor]
or property
```